Parboni Chattopadhyay

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Education

University of California, Irvine — Irvine, CA
B.S. Informatics (Human-Computer Interaction Specialization), Minor in Computer Science — *Jun 2026*

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C#, Java, SQL, HTML/CSS, PowerShell
AI & Data: Large Language Models (LLMs), LLM APIs, Prompt Engineering, Pandas, ETL Pipelines, Statistical Analysis
Frameworks & Tools: Angular, Node.js, Flask, Unity, Leaflet.js, Git, GitHub Actions, Figma
Systems & Security: Win32 API, Windows Security Tokens, Privacy-by-Design
Practices: SDLC, CI/CD, REST APIs, OOP, Code Review, Agile, Requirements Analysis, Usability Testing, WCAG 2.1

Experience

Software Engineer and LLM Product Manager (Contract) — Jul 2026 – Present
Systems Intelligence Health Equity Research Group, North Jersey Management Solutions — *New York, NY*
- Built a privacy-first desktop application that automates review of sensitive clinical documents for a healthcare research team, cutting review time 90%+ (from about 1 hour to a few minutes per document).
- Partnered with researchers to gather privacy and usability requirements, defined MVP scope, and shipped v1 as sole engineer and product manager, owning the full SDLC from design through testing and release.
- Applied privacy-by-design principles to meet strict privacy and security requirements, keeping sensitive data protected end to end.

Software Engineering Intern — Jun 2025 – Sep 2025
Numecent — *Irvine, CA*
- Engineered a C++ Win32 framework that runs PowerShell scripts across SYSTEM and USER security contexts in one workflow, enabling enterprise deployments that require both admin and per-user setup.
- Cut firewall-rule deployment time 98%+ (RealVNC 3.76s to 0.05s, AnyDesk 8.03s to 0.04s) with bulk PowerShell operations.
- Ensured reliable execution through security-token validation and privilege-aware CreateProcessAsUserW/LoadUserProfileW calls.
- Built PowerShell/JSON workflows to export and restore Windows users and groups with idempotent imports.

Software Engineering Grader — Sep 2024 – Mar 2026
University of California, Irvine — *Irvine, CA*
- Graded 200+ Python projects per quarter for Intro to Software Engineering, running unit tests and reviewing OOP design.

Research Experience

Undergraduate Research Software Engineer — Jun 2024 – Present
AI/LLM Education Impact Research, Donald Bren School of ICS — *University of California, Irvine*
- Built Python/Pandas ETL pipelines to analyze survey data on generative-AI use in software engineering education (ICCSE 2026).
- Scaled qualitative analysis of 1,200+ responses with LLM-assisted coding refined from historical codebooks.
- Automated processing of 14 open-ended survey questions via data dictionaries into reproducible codeword datasets.
- Linked AI usage to student satisfaction and academic-integrity concerns using one-hot encoding and Cramér's V.

Undergraduate Research Software Engineer — Jan 2026 – Jun 2026
Pediatric Sleep Research Lab, School of Medicine — *University of California, Irvine*
- Built a Unity/C# psychomotor vigilance task (PVT) game from clinical requirements; won 2nd Place nationally.
- Prototyped in 2D, then rebuilt as 3D basketball, golf, and tennis rounds for iPad.
- Refined age-appropriate interfaces with clinicians through iterative usability testing with 30+ users.

Systems Engineer — Aug 2025 – Dec 2025
NASA L'SPACE Research Program — *Remote*
- Led data visualization and analytics for a 10-person NASA mission proposal, creating its graphs and Appendix C3.
- Contributed mission architecture, requirements, and subsystem design through trade studies.

Projects

Zot Spot (VenusHacks 2025) | *Python, Flask, SQLite, JavaScript, Leaflet.js* | devpost.com/software/zot-spot — Feb 2025
- Built a real-time web app mapping free menstrual-product availability across UC Irvine restrooms with live stock reports.
- Added geolocation-based nearest-restroom search (Haversine distance) and responsive map and list views for desktop and mobile.

HealthHarmonies | *HTML/CSS, JavaScript, Figma* | parbonichattopadhyay.github.io/HealthHarmonies — Mar 2026
- Designed and developed a mobile-first healthcare resource app that improves access to health information for underserved users.
- Prototyped in Figma and built responsive interfaces using HCI principles, meeting WCAG 2.1 AA accessibility standards.

Awards & Publications

- **2nd Place (National)**, Capstone Design Conference, Texas A&M University; **3rd Place**, UC Irvine ICS Senior Design Expo
- **Co-author**, *Student Perceptions and Attitudes Toward AI in Software Engineering Education at an R1 University*, ICCSE 2026
- **Featured**, UCI ICS News (2026), for the pediatric sleep-assessment game